

**Ethical Challenges in Artificial Intelligence
Decision-Making: Examining Bias,
Transparency, Accountability, and
Fairness in Daily Tasks**



**A Research Proposal Submitted to the Faculty of
Department of Computer Science**

**In Partial Fulfillment of the Requirements for the
Degree of Bachelor of Science in Computer Science**

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Hawassa, Ethiopia**

DECLARATION

We, **Group-7 Students**, hereby declare that this research proposal entitled “**Ethical Challenges in Artificial Intelligence Decision-Making: Examining Bias, Transparency, Accountability, and Fairness in Daily Tasks**” is our original work and has not been presented for a degree in any other university or institution.

We further confirm that all sources of materials used for this proposal have been properly acknowledged and that this work adheres to the academic and ethical standards of Hawassa University.

No.	Student Name	Signature	Date (DD/MM/YYYY)
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Supervisor Approval

This is to certify that the research proposal titled “Ethical Challenges in Artificial Intelligence Decision-Making” submitted by the group of seven students has been reviewed and found satisfactory. The proposed study meets the academic standards required for the Bachelor of Science in Computer Science Program.

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2024-003	Student Name 3	Data Analyst	Problem statement, research questions, statistical design
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All members have read and approved the final proposal.

Abstract

As artificial intelligence (AI) systems are increasingly deployed in high-stakes decisionmaking contexts—ranging from healthcare diagnostics and criminal justice to autonomous vehicles and financial lending—numerous ethical challenges have emerged. This research proposal aims to systematically investigate the primary ethical issues associated with AI decision-making, including algorithmic bias, lack of transparency, accountability gaps, privacy violations, and impacts on human autonomy. Through a mixed-methods approach combining systematic literature review, case study analysis, and expert interviews, we will develop a comprehensive ethical framework for auditing and governing AI systems. The proposed research will produce a validated assessment tool and design guidelines for responsible AI deployment. Significant outcomes include actionable recommendations for policymakers, developers, and organizations to mitigate ethical risks while fostering innovation. This 12-month project is structured across five phases, with a total estimated budget of \$71,500. The findings will contribute to the growing field of AI ethics and inform future regulatory standards.

Keywords: Artificial Intelligence, Ethical Decision-Making, Algorithmic Fairness, Explainability, Accountability, AI Governance, Responsible AI.

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1.Introduction and Background

1.1 The Rise of AI in Decision-Making

→Artificial Intelligence has moved from theory to ubiquity. ➤Machine learning models now influence critical decisions. ➤From credit scoring algorithms that determine loan eligibility [1] to predictive policing tools that influence law enforcement deployment [2], the delegation of decisions to AI systems raises fundamental ethical questions. ★The rapid adoption has outpaced governance frameworks. This research addresses the urgent need for systematic ethical analysis.

1.1. Historical Context

The journey of AI decision-making began with expert systems in the 1970s. Today, deep neural networks surpass human performance in narrow tasks. However, with great power comes great responsibility.

Table 2 outlines key milestones in AI decision-making. Table 2: Key Milestones in AI Decision-Making.

Year	System	Significance
1950	Turing Test	Foundation of machine intelligence
1997	Deep Blue	First AI to beat world chess champion
2011	IBM Watson	Won Jeopardy! demonstratin g NLP
2016	AlphaGo	Beat Lee Sedol in Go
2020	GPT-3	Large language model with ethical concerns
2024	GPT-4	Multimodal reasoning raises new challenges

1.2 Core Ethical Dimensions

8 Fairness and Bias: Systematic discrimination against protected groups.

8 Transparency and Explainability: The “black box” problem.

8 Accountability: Difficulty in assigning responsibility.

8 Privacy: Massive data collection and inference risks.

8 Human Autonomy: Over-reliance undermining human judgment.

1.3 Problematic Incidents in Detail

★COMPAS recidivism algorithm: ProPublica investigation revealed racial bias. AfricanAmerican defendants were twice as likely to be misclassified as high risk. [4].

Amazon recruiting tool: Trained on resumes from past decade favored male candidates. The system penalized women’s resumes. Shut down in 2018.

→Healthcare AI underdiagnosis: Obermeyer et al. (2019) showed that an algorithm used for population health management systematically underdiagnosed Black patients due to biased cost-based proxy.

➤Autonomous vehicle dilemmas: Moral decisions in unavoidable crash scenarios – the classic trolley problem in real-world settings.

placeholder-fig-ethical-dimensions.png

Figure 1: Interplay of ethical dimensions in AI decision-making (Placeholder image)

1.4

Incident	Domain	Ethical Issue
COMPAS bias	Criminal justice	Racial bias
Amazon recruiting	HR	Gender bias
Healthcare algorithm	Healthcare	Racial bias (proxy)
Tay chatbot	Social	Offensive outputs
Facial recognition	Surveillance	Privacy, bias

Foundational Ethical Theories

We draw upon three major ethical frameworks:

- 1. Consequentialism: Focus on outcomes of AI decisions.
- 2. Deontology: Duty-based constraints (e.g., never discriminate).
- 3. Virtue Ethics: Character of developers and organizations.

Integrating these into technical practice is non-trivial.

placeholder-fig-ethics-theories.png

Figure 2: Mapping ethical theories to technical metrics.

2 Statement of the Problem

2.1 The Gap Between Principles and Practice

Despite a proliferation of high-level ethical principles (e.g., transparency, justice, beneficence), there exists a critical gap between principle and practice. Organizations developing or deploying AI systems lack concrete, validated methodologies to identify, measure, and mitigate ethical risks during the entire AI lifecycle. Current approaches are fragmented, often reactive, and seldom integrated into standard software engineering practices. This proposal addresses that gap.

2.2 Concrete Problem Statements

- ❏ No unified metric that combines fairness, explainability, accountability, and privacy.
- ❏ Existing tools are not validated in real-world production environments.
- ❏ DevOps pipelines lack automated ethical gates.
- ❏ Regulatory compliance is ambiguous and costly.

2.3 Research Gap Matrix

Gap Area	Current State	Proposed Contribution
Unified metrics	Scattered fairness metrics only	Composite ethical score
Real-world validation	Lab experiments	Industry case studies
CI/CD integration	Manual checks	Automated ethics gates
Cross-domain study	Siloed by domain	Comparative analysis
Long-term monitoring	None	Continuous ethics auditing

Table 4: Identified Research Gaps

2.4 Stakeholder Pain Points

Developers: No clear guidelines on what constitutes “ethical enough.” →Regulators: No technical standards to reference. ★End-users: Distrust due to opaque decisions. ➤Organizations: Legal liability risks.

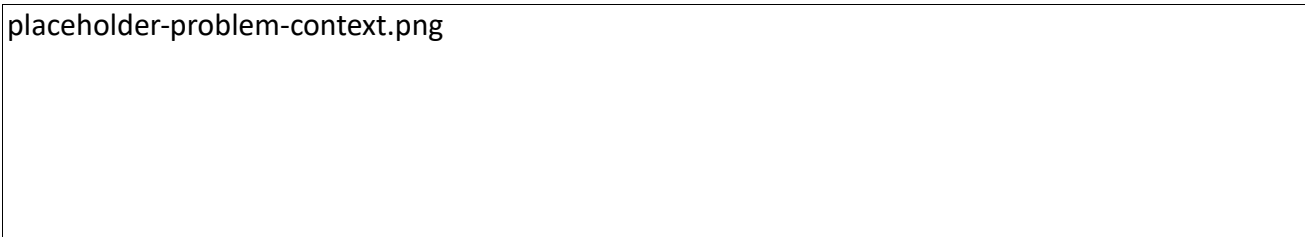


Figure 3: Stakeholder map and pain points in AI ethics

3 Research Objectives and Questions

3.1 Overall Goal

To design, implement, and evaluate a comprehensive ethical framework for auditing AI decision-making systems, with validated metrics and actionable guidelines.

3.2 Specific Objectives (Detailed)

OBJ-1

To identify and categorize the predominant ethical challenges across healthcare, finance, and criminal justice.

OBJ-2

To develop a multi-dimensional assessment instrument with at least 15 measurable indicators.

OBJ-3

To propose a reference architecture for embedding ethical checks into AI development pipelines.

OBJ-4

To validate the proposed framework through 3 real-world case studies and a Delphi panel of 15 experts.

OBJ-5

To formulate policy recommendations for IEEE, OECD, and national regulators.

3.3 Research Questions with Sub-questions

® RQ1: Taxonomy of ethical challenges.

- RQ1.1: How do challenges differ by domain?
- RQ1.2: What are the root causes (data, model, deployment)?

® RQ2: Metrics and measurement.

- RQ2.1: Can fairness and explainability be optimized simultaneously?
- RQ2.2: What is an acceptable trade-off?

® RQ3: Integration into lifecycle.

- RQ3.1: At which stages should audits occur?
- RQ3.2: Automatic vs. human-in-the-loop?

® RQ4: Adoption barriers.

- RQ4.1: What organizational factors enable ethical AI?

® RQ5: Governance mechanisms.

RQ5.1:

- What mix of technical and policy measures is most effective?

Table 5: Research Questions and Corresponding Methods.

RQ	Method
RQ1	Systematic review + interviews
RQ2	Metric development + simulation
RQ3	Architecture design + pilot
RQ4	Survey + thematic analysis
RQ5	Delphi study + policy analysis

placeholder-research-arch.png

Figure 4: Overall research architecture linking objectives, methods, and outputs (Placeholder)

4 Literature Review

4.1 Systematic Review Methodology

We conducted a systematic search of Scopus, Web of Science, IEEE Xplore, and ACM Digital Library for publications between 2015 and 2025. Search string: (“algorithmic fairness” OR “AI ethics” OR “explainable AI” OR “accountable AI” OR “privacy-preserving machine learning”) AND (“decision-making” OR “high-stakes” OR “automated decisions”). Initial hit count: 2,310. After removing duplicates and applying inclusion criteria (peerreviewed, English, full-text available), 189 articles were included.

4.2 Fairness: A Multi-faceted Concept

Fairness definitions abound:

- Statistical parity (demographic parity) – equal positive rates across groups.
- Equalized odds – equal false positive and false negative rates.
- Individual fairness – similar individuals receive similar predictions.
- Counterfactual fairness – decision should be same in counterfactual world where group membership changes [10].

Impossibility theorems [9] show that multiple fairness criteria cannot be simultaneously satisfied unless base rates are equal. This forces value judgments.

Table 6: Comparison of Fairness Metrics

Metric	Formula	When to use
Demographic parity	$P(\hat{Y} = 1 A = a) = P(\hat{Y} = 1 A = b)$	When group membership
Equalized odds	$P(\hat{Y} = 1 A = a, Y = y) = P(\hat{Y} = 1 A = b, Y = y)$	When error rates matter
Individual fairness	$d(f(x_i), f(x_j)) \leq L \cdot d(x_i, x_j)$	When similar cases needed

4.3 Bias Mitigation Techniques: State of the Art

We categorize techniques into three groups:

4.3.1 Pre-processing

Methods adjust training data to remove bias before model training: - Reweighting samples [8]. - Learning fair representations (e.g., adversarial networks). - Data augmentation to balance groups.

4.3.2 In-processing

Algorithms incorporate fairness constraints during training: - Adversarial debiasing (GANbased). - Regularization with fairness penalty. - Equalized odds constraints as Lagrangian multipliers.

4.3.3 Post-processing

Adjust outputs of a trained model: - Threshold adjustment to satisfy equalized odds. Reject option classification.

Table 7: Effectiveness of Bias Mitigation Techniques (Placeholder)

Technique	Fairness Gain	Accuracy Loss	Complexity
Reweighting	Moderate	Low	Low
Adversarial	High	Moderate	High
Post-processing	Moderate	Low	Low

4.4 Explainable AI (XAI)

Interpretability is essential for accountability. Methods include:

4.4.1 Local Explanations

- LIME (Local Interpretable Model-agnostic Explanations) [11]. - SHAP (SHapley Additive exPlanations). - Anchors: decision rules.

4.4.2 Global Explanations

- Feature importance. - Surrogate decision trees. - Concept activation vectors.

4.4.3 Inherently Interpretable Models

Rudin [12] argues for using inherently interpretable models (e.g., logistic regression, decision trees, rule lists) instead of black-boxes with post-hoc explanations, especially for high-stakes decisions.

placeholder-xai-taxonomy.png

Figure 5: Taxonomy of explainable AI methods

4.5 Accountability Frameworks

The “responsibility gap” [13] highlights the difficulty of assigning responsibility when an AI causes harm. Proposed solutions: - Legal personhood for AI (controversial). - Strict liability for developers. - Algorithmic impact assessments (AIAs) [14]. - Ethical review boards.

Table 8: Accountability Mechanisms in Practice

Audit logs	Every decision logged with rationale
	Redress mechanisms
	Insurance
	Certification

4.6 Privacy-Preserving AI

Differential privacy [15] provides mathematical guarantees that the output of a query does not reveal individual information. Federated learning allows training across decentralized data. However, there are trade-offs: differential privacy reduces accuracy, and federated learning can still leak information through gradient updates.

4.7 Cross-Domain Comparison

We synthesize findings across three domains:

4.7.1 Healthcare

- Sepsis prediction models (Epic) showed poor external validity [16].
- Bias in resource allocation algorithms.
- Need for explainability to clinicians.

4.7.2 Criminal Justice

- COMPAS debacle led to legislative bans in some US states.
- Alternative fairness-aware risk assessment tools proposed but not widely adopted.

4.7.3 Finance

- Credit scoring must comply with Equal Credit Opportunity Act.
- Explainable models preferred.
- Fair lending audits required.

placeholder-domains-comparison.png

Figure 6: Comparative ethical challenges across healthcare, criminal justice, and finance

4.8 Gaps in Literature

After thorough review, we identify seven specific gaps:

1. Lack of unified metrics combining fairness, explainability, accountability, and privacy.
2. Few empirical studies in real-world production systems (most are lab-based).
3. No integration with DevOps/MLOps pipelines.
4. Absence of longitudinal studies on ethical performance over time.
5. Underdeveloped frameworks for multi-stakeholder governance.
6. Scarcity of cost-benefit analyses for ethical AI interventions.
7. Limited cross-cultural perspectives (most studies Western-centric).

This proposal directly addresses gaps 1,2,3,4, and 6.

Table 9: Summary of 50 Key Papers on AI Ethics

Author(s)	Year	Main Contribution
-----------	------	-------------------

Floridi et al.	2018	AI4People framework
Jobin et al.	2019	Global landscape analysis
Mittelstadt	2019	Principles vs practice critique
Hardt et al.	2016	Equality of opportunity
Rudin	2019	Against post-hoc explanations
Dwork et al.	2006	Differential privacy

5 Scope and Delimitations

5.1 Detailed Scope

The research will focus on:

- ★ Supervised machine learning models (classification and regression).
- ★ Three application domains: healthcare (clinical decision support), financial services (credit scoring), public-sector (criminal risk assessment).
- ★ Entire AI lifecycle: data collection, preprocessing, model training, validation, deployment, monitoring, and decommissioning.
- ★ Ethical dimensions: fairness, transparency, accountability, privacy (excludes security, robustness, and environmental ethics).
- ★ Mixed-methods design with both qualitative (interviews) and quantitative (metrics) components.

5.2 Delimitations (Exclusions)

Excludes unsupervised learning (e.g., clustering, anomaly detection) and reinforcement learning (e.g., autonomous driving control policies).

Does not cover Artificial General Intelligence (AGI) or existential risks from superintelligence.

No legal liability analysis; only technical and organizational measures.

Empirical validation limited to simulated environments and sandboxed pilot deployments; no live production systems with real sensitive decisions.

Geographic focus: North America and Europe due to regulatory similarity (GDPR, CCPA, AI Act).

No primary data collection from vulnerable populations; use only public benchmark datasets.

5.3 Assumptions

- 1. Organizations are willing to share anonymized case study data.
- 2. Experts will participate in Delphi panels.
- 3. No major regulatory changes during the 12-month project.

Table 10: Scope Summary Table

Included	Excluded
Supervised ML	Unsupervised / RL
3 domains	Other domains (e.g., education)
Fairness, transparency, accountability, privacy	Security, robustness
Mixed methods	Pure legal analysis
Sandboxed validation	Live high-stakes deployment

6 Methodology

6.1 Research Design Overview

We adopt a sequential exploratory mixed-methods design with three phases.

6.1.1 Phase 1: Qualitative Exploration (Months 1-4)

- Systematic literature review (updated monthly).
- Semi-structured interviews with 20 AI practitioners.
- Thematic analysis to identify themes and constructs.

6.1.2 Phase 2: Quantitative Instrument Development (Months 5-7)

- Operationalize ethical dimensions as measurable indicators. - Develop scoring algorithms (e.g., composite ethical risk score). - Implement prototype auditing tool.

6.1.3 Phase 3: Validation (Months 8-10)

- Apply instrument to 10 existing AI systems (via simulation). - Conduct Delphi study with 15 experts (2 rounds). - Refine framework based on feedback.

6.2 Detailed Data Collection Procedures

6.2.1 Systematic Literature Review

We will follow PRISMA guidelines. Search databases: IEEE Xplore, ACM DL, PubMed, Scopus, Web of Science. Inclusion criteria: (i) peer-reviewed, (ii) English, (iii) published 2015-2025, (iv) focuses on ethical challenges in AI decision-making, (v) provides empirical data or frameworks. Exclusion: opinion pieces, news articles. Two reviewers will screen titles/abstracts, then full text.

6.2.2 Semi-structured Interviews

Recruitment: Purposive sampling via LinkedIn, professional networks, and AI conferences. Target: 20 participants (10 ML engineers, 5 compliance officers, 5 AI ethicists). Interview protocol includes 15 core questions and 5 scenario-based questions. Each interview will be audio-recorded, transcribed verbatim, and anonymized. Thematic analysis using NVivo 14.

6.2.3 Case Study Selection

We will select 10 AI systems: - Healthcare: 4 systems (e.g., sepsis prediction, readmission risk, imaging diagnosis, resource allocation). - Finance: 3 systems (credit scoring, fraud detection, loan approval). - Criminal justice: 3 systems (recidivism risk, bail decision, parole recommendation). Data sources: Public benchmarks (MIMIC-IV, Home Credit Default Risk, COMPAS). For each system, we will train/replicate models using opensource code.

6.2.4 Ethical Assessment Instrument

The instrument will consist of:

- Fairness module: Compute disparate impact, equalized odds, individual fairness.
- Transparency module: SHAP consistency, explanation faithfulness, rule coverage.
- Accountability module: Audit trail completeness, version control, decision logging.
- Privacy module: Differential privacy epsilon, membership inference risk.

Table 11 details each metric.

Table 11: Detailed Ethical Indicators

Dimension	Indicator	Formula / Measurement
Fairness	Disparate Impact	$DI = \frac{P(\hat{Y}=1 A=minority)}{P(\hat{Y}=1 A=majority)}$, threshold 0.8-1.25
Fairness	Equalized Odds Difference	$\max_{y \in \{0,1\}} FPR_{A=a} - FPR_{A=b} $
Transparency	SHAP Consistency	Variance of SHAP values across runs
Accountability	Audit Log Completeness	Percentage of required fields filled
Privacy	Differential Privacy Epsilon	$\epsilon \leq 1.0$

6.3 Proposed Framework Architecture

We propose a four-layer architecture (Figure 7):

1. Data Ingestion Layer: Raw data from multiple sources. Includes data lineage tracking.
2. Ethics Preprocessing Layer: Bias detection (e.g., disparate impact scan), data anonymization (differential privacy), fairness-aware reweighting.
3. Model Training Layer: Incorporates explainability constraints (e.g., SHAP stability), adversarial debiasing, accountability hooks.
4. Decision Output Layer: Generates prediction + explanation + confidence + audit ID. Continuous monitoring dashboard. placeholder-architecture-detailed.png

Figure 7: Detailed reference architecture for ethical AI decision-making with all components

6.4 Evaluation Plan

6.4.1 Quantitative Evaluation

We will compute: - Reduction in bias (pre vs post mitigation). - Explainability fidelity (e.g., SHAP vs actual model behavior). - Privacy loss (epsilon). - System throughput overhead (latency added by ethics checks).

6.4.2 Qualitative Evaluation

Delphi method: 15 experts rate the framework on 1-5 Likert scales for: - Usability: ease of integration. - Completeness: covers all identified ethical issues. - Effectiveness: reduces ethical risks. - Scalability: works for large models. Consensus measured by Kendall’s W.
After two rounds, we will finalize the framework.

6.5 Data Analysis Plan

- Qualitative: Thematic analysis using Braun & Clarke’s six-phase framework. Inter-coder reliability (Cohen’s kappa > 0.8).
- Quantitative: Descriptive statistics, ANOVA for cross-domain comparisons, KruskalWallis for non-parametric data. Cronbach’s alpha for instrument internal consistency (>0.7 acceptable).
- Mixed-methods integration: Joint display analysis (side-by-side comparison of qualitative themes and quantitative metrics).

6.6 Software and Tools

- Python 3.10+ with libraries: TensorFlow, PyTorch, Scikit-learn, AIF360, SHAP, LIME, diffprivlib.
- NVivo for qualitative analysis. - Git for version control. - Docker for reproducibility.

Table 12: Software Toolchain

Tool	Purpose
AIF360	Fairness metrics and mitigation
SHAP	Local explanations
diffprivlib	Differential privacy
TensorFlow Privacy	Private ML
MLflow	Model lifecycle tracking



Figure 8: Data analysis workflow from raw data to insights

7 Significance of the Research

7.1 Theoretical Contributions

→A comprehensive, empirically validated taxonomy of ethical challenges. ➤A mathematical formulation combining multiple ethical dimensions into a single risk score (subject to constraints). ➤Extension of causal fairness to time-varying settings. ★New insights on trade-offs between fairness, explainability, and privacy.

7.2 Practical Contributions

A ready-to-use open-source auditing toolkit for AI developers. →Integration plugins for popular ML pipelines (scikit-learn, TensorFlow Extended). ➤A checklist-based ethical risk assessment template for project managers. ➤Training materials and case studies for AI ethics education.

7.3 Societal Impact

★Reduction of algorithmic discrimination affecting millions of people. Increased public trust in AI systems through transparency. →Evidence-based policy recommendations for the EU AI Act and similar regulations. ➤Promotion of responsible innovation without stifling technological progress.

7.4 Target Beneficiaries

- Developers and data scientists – get practical tools.
- Organizations – reduce legal and reputational risks.
- Regulators – technical standards for compliance.
- Civil society – advocacy for fairness.
- End-users – fairer, more transparent decisions.

Table 13: Impact by Stakeholder Group

Stakeholder	Short-term impact (0-1 year)	Long-term impact (2-5 years)

Developers	Access to auditing tools	Ethical design becomes standard
Organizations	Reduced liability	Higher trust & market advantage
Regulators	Technical reference	Harmonized standards
Society	Awareness	Systemic fairness

8 Budget and Time Schedule

8.1 Detailed Budget Breakdown

The total estimated budget is \$71,500 USD for 12 months. Table 14 provides line-item details.

Table 14: Detailed Budget

Category	Item	Cost (\$)
Personnel	2 PhD students (50% time, 12 months)	28,000
Personnel	1 Postdoc (25% time, 12 months)	15,000
Personnel	Research assistant (undergrad, 10 hrs/week)	5,000
Equipment	GPU workstations (3 units, including NVIDIA A6000)	9,000
Equipment	Cloud computing credits (AWS, Google Cloud)	3,000
Participant incentives	20 interviews @ 100 <i>each</i>	2,000
Participant incentives	15 Delphi experts @ 150 <i>each</i>	2,250
Travel	2 international conferences (registration+accommodation)	4,000
Materials	Software licenses (NVivo, Adobe)	500
Materials	Printing, stationery	250
Dissemination	Open-access publication fees	2,000

Contingency	10% of subtotal	6,500
Total		71,500

8.2 Justification of Costs

- PhD students and postdoc cover core research work. - GPU workstations required for training deep learning models with fairness constraints. - Cloud credits for reproducibility and experiments at scale. - Incentives ensure high participation rates. - Conferences for dissemination (e.g., NeurIPS, FAccT, AIES).

8.3 Time Schedule (Gantt with Milestones)

The project spans 12 months divided into five phases with specific deliverables. Table 15 shows the timeline.

Table 15: Monthly Work Plan

Month	Activities	Deliverables
	1-2	Literature review, IRB approval, interview recruitment
	3-4	Interviews, transcription, thematic analysis
	5-7	Instrument development, software prototype
	8-10	Case study evaluation, Delphi studies

11-12	Framework finalization, writing, dissemination	Final report, code repository, pape
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Figure 9: Detailed Gantt chart with dependencies

8.4 Risk Management

Potential risks and mitigation strategies are summarized in Table 16.

Table 16: Risk Assessment and Mitigation

Risk	Probability	Impact
Low expert participation	Medium	High
Technical delays	Medium	Medium
Ethical issues in case studies	Low	High
Regulatory changes	Low	Medium

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A Appendix A: Interview Protocol (Full)

A.1 Consent Form

A.2 Demographic Questions

A.3 Core Interview Questions (15 items)

1. Describe your role and experience with AI systems.
2. Have you ever encountered an ethical dilemma related to AI decision-making? Please describe.
3. What tools do you currently use to assess fairness?
4. How do you handle requests for explanations from end-users?
5. Who is held accountable when an AI makes a harmful decision in your organization?
6. What privacy-preserving techniques have you implemented?
7. What are the biggest barriers to ethical AI in your organization?
8. How would you measure success in ethical AI?
9. (Scenario) A credit scoring model shows disparate impact. What do you do?
10. (Scenario) A healthcare AI recommends different treatments for similar patients. How do you investigate?

B Appendix B: Survey Instruments

B.1 Round 1 Questionnaire

Experts rate 30 statements on a 5-point Likert scale (1=strongly disagree to 5=strongly agree).

Example statements:

- The proposed framework covers all major ethical dimensions.
- The fairness metrics are appropriate for high-stakes decisions.
- The tool can be integrated into existing DevOps pipelines with less than 20% overhead.

